

# What is Software Testing?

## Definition (Interview)

**Software Testing is the process of checking whether a software application works as expected and finding bugs before it is delivered to users.**

## Easy Words

Imagine you built a calculator.

You test:

$2 + 2 = 4$  ✓✓

$2 + 2 = 5$  ✗

If something is wrong, that's a **bug**.

Your job as a QA Engineer is to find bugs before customers do.

## Why is Software Testing Important?

There are **4 main reasons**.

- **Quality**
  - The software should work correctly.
  - Example: Instagram should upload photos without crashing.
- **Security**
  - Protect user data.
  - Example: Hackers should not be able to steal passwords.
- **Save Money**
  - Finding bugs early is cheaper.
  - Example: Fixing a login bug before release takes minutes.
  - Fixing it after 10 million users complain costs a lot more.
- **Customer Satisfaction**
  - Users expect apps to work smoothly.
  - Happy users = successful product.

## Important Words

- **Bug**
  - Problem found during testing.
  - Example: Login button doesn't work.
- **Defect**

- Difference between expected and actual result.
- Expected: Password should be hidden.
- Actual: Password is visible.
- That's a defect.
- **Failure**
  - When software completely fails.
  - Example: Website crashes.

## **Software Testing Categories**

You only need to know these today.

- **Functional Testing**
  - Question: "Does it work?"
  - Example: Login, Search, Checkout and Payment
- **Non-Functional Testing**
  - Question: "How well does it work?"
  - Example:
    - Is it fast?
    - Is it secure?
    - Can 1000 users use it together?
- **Maintenance Testing**
  - Testing after updates.
  - Example: Developer fixes Login.
  - QA checks nothing else broke.

## **Levels of Testing**

Remember this order.

Unit -> Integration -> System -> User Acceptance Testing (UAT)

- **Unit**
  - Smallest code.
  - Done by Developer.
- **Integration**
  - Modules working together.
  - Example: Login + Database
- **System**
  - Whole application.
  - Example: Complete Amazon website.
- **UAT**
  - Customer checks everything before release.

## **Functional Testing Types (Just Remember the Names)**

- Unit
- Integration
- System
- UAT
- Smoke
- Sanity
- API
- Database

## **Non-Functional Testing Types**

Only remember these:

- Performance
- Security
- Usability
- Compatibility

## **Real-Life Example**

Imagine you're testing Foodpanda.

Functional Testing:

- Can I log in?
- Can I search food?
- Can I place an order?

Non-Functional Testing:

- Does the app load quickly?
- Does it crash?
- Is my payment secure?

## **Severity = Damage**

How serious is the problem?

## **Priority = Deadline**

How quickly should we fix it?

This is a realistic interview scenario.

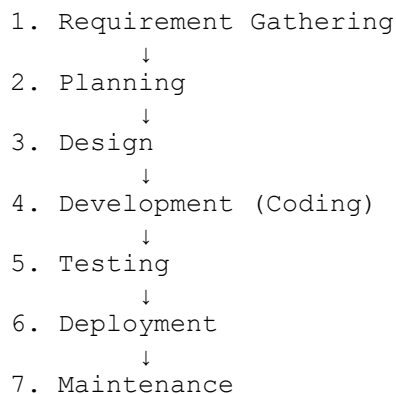
## SDLC vs STLC

### What is SDLC?

**SDLC = Software Development Life Cycle**

It is the **complete process of building software** from the beginning until it is maintained after release.

### Steps of SDLC



### Example

Imagine a company wants to build **Foodpanda**.

The team first asks:

- What features do customers want?
- What should the app look like?
- Developers write the code.
- QA tests it.
- The app is launched.
- Bugs are fixed later.

This whole journey is **SDLC**.

### What is STLC?

**STLC = Software Testing Life Cycle**

STLC is **only the testing process**. QA Engineers mainly follow STLC.

### Steps of STLC

1. Requirement Analysis
- ↓
2. Test Planning
- ↓
3. Test Case Development
- ↓
4. Test Environment Setup
- ↓
5. Test Execution
- ↓
6. Test Closure

## Example

The developers finish the Login page. Now QA starts.

QA:

- Reads the requirements.
- Makes a testing plan.
- Writes test cases.
- Sets up the testing environment.
- Tests the Login page.
- Creates a testing report.

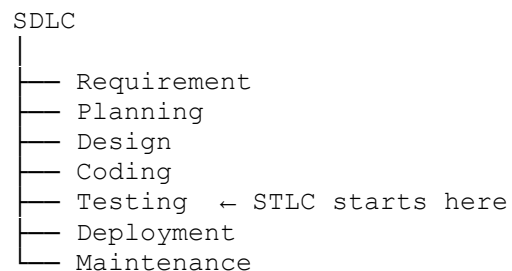
This is **STLC**.

## Easy Way to Remember

**SDLC:** Build the software.

**STLC:** Test the software.

## Relationship



Notice:

Testing is one phase of SDLC. Inside that testing phase, QA follows **STLC**.

## Side-by-Side Comparison

| <b>SDLC</b>                                                    | <b>STLC</b>                           |
|----------------------------------------------------------------|---------------------------------------|
| Software Development Life Cycle                                | Software Testing Life Cycle           |
| Used to develop software                                       | Used to test software                 |
| Followed by Developers, QA, Designers, Business Analysts, etc. | Mainly followed by QA/Testers         |
| Starts from requirements                                       | Starts after QA receives requirements |
| Ends with maintenance                                          | Ends with test closure                |
| Goal: Build software                                           | Goal: Ensure software quality         |

## Real-Life Example

Suppose you're building an Online Shopping App.

### **SDLC**

Business says:

"We need an online shopping app."

↓

UI/UX designer creates screens.

↓

Developer writes code.

↓

QA tests the app.

↓

App is released.

↓

Later, bugs are fixed.

This whole process is **SDLC**.

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## **STLC**

QA receives the Login feature.



Reads requirements.



Writes test cases.



Tests Login.



Finds bugs.



Reports bugs.



Retests after fixes.



Testing completed.

This is **STLC**.

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## **Interview Answer**

### **What is SDLC?**

SDLC (Software Development Life Cycle) is the complete process of developing software, from gathering requirements to maintenance after deployment.

### **What is STLC?**

STLC (Software Testing Life Cycle) is the process followed by QA engineers to plan, design, execute, and complete software testing.

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### **Most Asked Interview Question**

**Q: What is the difference between SDLC and STLC?**

A good answer is:

SDLC is the complete software development process, while STLC is only the software testing process. Testing is one phase of SDLC, and during that phase QA engineers follow STLC.

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